

Week 04: Transformers without heavy mathematics

Generative AI | MSc teaching notes and further reading | 2027

Learning target

Explain tokens, attention weights, masking and model roles.

Context and teaching plan

A document assistant must choose what earlier words matter when drafting a sentence.

Contact time	Suggested use
Two hours: concepts	Case introduction, four ideas, worked example and misconception.
Two hours: practice	10 min recall, 20 min demo, 40 min guided practice, 25 min independent change, 15 min review, 10 min exit check.

Explain the key ideas

Token

A model processes pieces of text represented as IDs. A word may use more than one token.

Teacher question: Where does token appear in the weekly case, and what could go wrong if we omit it?

Attention

Each query uses scores to weight available tokens; the weighted combination carries contextual information.

Teacher question: Where does attention appear in the weekly case, and what could go wrong if we omit it?

Mask

A decoder mask prevents a token from using future tokens while generating left to right.

Teacher question: Where does mask appear in the weekly case, and what could go wrong if we omit it?

Model roles

Encoders read inputs; decoders generate; encoder-decoder models read then generate from a separate component.

Teacher question: Where does model roles appear in the weekly case, and what could go wrong if we omit it?

Worked example and answer guide

- Input: Scores for three tokens: 1, 2, 0.
- Change or question: Softmax weights approximately 0.245, 0.665, 0.090.
- Supported observation: If the third token is future text, its masked weight becomes 0.

Reasoning: state the given evidence first, then the calculation or rule, then the conclusion. If a value is absent, say what cannot be concluded. Keep any simulated mechanism distinct from a trained model or deployed service.

Teach the week as a story

The confusing sentence: In “Ava told Bea that she won,” which earlier word does “she” refer to? A model must use context, yet ambiguity can remain.

Ask for a quick prediction before each technical explanation. Return to the prediction after the example and ask what evidence changed the answer. The analogies below are memory aids, and each has a stated limit.

Token: analogy, mechanism and check

Familiar picture: A sentence is cut into tiles before a game reads it.

Familiar part	Technical meaning
Sentence	input string
Tiles	tokens
Tile numbers	IDs

Where this analogy breaks: Token pieces need not be complete words; the vocabulary is model-specific.

Actual idea: A model processes pieces of text represented as IDs. A word may use more than one token.

Work it aloud: “unbelievable” might be one piece or several; record the tokenizer used before counting.

Ask before revealing: Why can word count differ from token count?

Supported response: A tokenizer may split a word into several pieces.

Look for this misconception: One word always equals one token.

Attention: analogy, mechanism and check

Familiar picture: A reader highlights earlier words that matter to the current word.

Familiar part	Technical meaning
Current word	query
Highlight strength	attention weight
Notes	weighted context

Where this analogy breaks: Attention weights alone are not a complete explanation of model decisions.

Actual idea: Each query uses scores to weight available tokens; the weighted combination carries contextual information.

Work it aloud: Scores [1,2,0] yield about [0.245,0.665,0.090]; the middle token gets most weight.

Ask before revealing: Do weights identify a proven causal explanation?

Supported response: No. They show one computation, not the whole model's reason.

Look for this misconception: The largest attention cell proves the final answer.

Mask: analogy, mechanism and check

Familiar picture: An exam candidate may consult earlier pages, not tomorrow's answer sheet.

Familiar part	Technical meaning
Earlier page	past token
Covered page	masked future token
Allowed view	causal context

Where this analogy breaks: A mask is an enforced attention constraint; it does not ensure factual accuracy.

Actual idea: A decoder mask prevents a token from using future tokens while generating left to right.

Work it aloud: Mask the third of three scores [1,2,0]; renormalised weights are about [0.269,0.731,0].

Ask before revealing: What happens to the forbidden position?

Supported response: Its weight is zero; visible positions are renormalised.

Look for this misconception: Masking deletes a token from the vocabulary.

Model roles: analogy, mechanism and check

Familiar picture: A reader, a writer, and a reader-writer have different jobs.

Familiar part	Technical meaning
Reader	encoder
Writer	decoder

Where this analogy breaks: Many current systems combine components; this is a useful role sketch, not a full taxonomy.

Actual idea: Encoders read inputs; decoders generate; encoder-decoder models read then generate from a separate component.

Work it aloud: Classify a sentence: encoder. Continue it: decoder. Translate it: encoder-decoder.

Ask before revealing: Which role suits translation in this simplified comparison?

Supported response: An encoder reads source text and a decoder writes target text.

Look for this misconception: An encoder always generates the final long reply.

New situation for discussion

A drafting decoder has the next word hidden. What happens to that column in its attention grid?

Memory cue: Tokenise, score visible context, and choose the model role for reading or writing.

Guided laboratory

Compute attention for three tokens and explain one mask.

1. Tokenize a tiny three-word classroom example
2. Compute normalized attention weights from simple scores
3. Apply a mask to hide a future word
4. Compare an encoder and a decoder task

Individual change: Mask the last token and recompute the weights. Expected evidence: Annotated attention table and one model-role choice.

Checks and common error

Measure: Weights sum to one over visible tokens. Record the input, setting, source version, result and the relevant numerator and denominator.

Plausible incorrect claim: Attention weights are a complete explanation of a model decision.

Correction: They show one internal weighting operation; they do not prove causal importance or factual correctness.

If learners disagree, reproduce the same case and compare the underlying source or expected answer. Record unresolved cases for instructor review.

Further discussion

- How could the conclusion change if the source or input changes?
- Which part of the result is observed and which part is an assumption?
- What simple test would detect the failure described above?

References and optional extension

Core readings are official vendor documentation or original research. Read the relevant section; an account or paid API is unnecessary for this notebook. Page details may change before 2027.

Vaswani et al., Attention Is All You Need. <https://arxiv.org/abs/1706.03762>

Google, Transformer model overview. <https://cloud.google.com/discover/what-is-gpt>

Optional extension: replace one classroom proxy with a locally approved model or service, keep the original baseline, repeat the same checks, and state the additional privacy and cost assumptions.

Curriculum scope: the supplied 12-week teaching sequence, recorded in docs/SYLLABUS_ALIGNMENT.md. The original module descriptor is retained in sources/originals/07-Descriptor-Generative-AI-new-.docx. Supporting project notes: Deep-research.txt, RAG.txt, AI-Agents.txt, MCP-&-Vector-Database.txt, Multi-Agent-AI-System.txt, Eval-&-Finetune&-Loop-Engineering.txt. These are planning sources, not empirical performance evidence.

Student revision and exam preparation

Outcomes: LO1, LO2 | 2 hours concepts + 2 hours guided practice

The idea in plain language

Tokens are model-specific pieces of input, not always words. Attention combines representations using weights. A causal mask stops a decoder using future positions. Encoders suit understanding inputs; decoders generate continuations; encoder-decoder models read an input and generate a corresponding output.

A worked example

Weights 0.25, 0.65 and 0.10 sum to 1. A token in position two cannot see position three under a causal mask. Scores [1,2,-infinity] produce weights about [0.269,0.731,0]. A large attention weight alone does not establish causal explanation or factual accuracy.

Practice before reading the answers

1. Why does a decoder use a causal mask? (4 suggested practice marks)

2. For weights 0.25, 0.65, 0.10, which input receives most weight? (4 suggested practice marks)
3. Choose a model role for translating one sentence into another language. (4 suggested practice marks)

Draw the main diagram from memory, explain each arrow, then check it against the slides. Answer the three questions without notes before reading the model answers.

Model answers and marking guidance

1. To ensure generation at a position depends only on permitted earlier/current positions, not future target tokens.
2. The second input, with 0.65. The three weights sum to 1.
3. An encoder-decoder design is a suitable example: read the source then generate the target. Other architectures can also translate.

For each four-mark practice answer: two marks for the correct idea, one for a relevant example or calculation, one for an appropriate explanation or limitation. These suggested practice marks do not change the module assessment scheme.

Official reading and video support

G1 Google: Introduction to Generative AI

https://www.cloudskillsboost.google/course_templates/536

Study focus: Generation versus traditional machine learning.

G5 Google Cloud Tech: Attention mechanism: Overview

<https://www.youtube.com/watch?v=fjJOgb-E41w>

Study focus: Attention intuition. Verified channel in indexed metadata; full transcript not retrieved.

Selected source concepts inform the lesson. Examples, figures, questions and solutions are original classroom material; no exam questions are claimed to be official.